数理分析方法

实数、公理与完备性

课件编号：105-ZSM-PPT　原始 PPT：67 页

逐页保留原始 PPT 画面 + 中文零基础自学讲解 + 证明专题

阅读说明：本讲义每一页原课件均以原始 PDF 页面完整嵌入；其后的中文解读旨在帮助初学者识别定义、假设、论证步骤。章末为从零展开的核心证明与复习提示。讲义属于学习辅助材料，不代替原教师课件。参考：Gérard Debreu (1959), *Theory of Value*。

目录

1 学前导读

这一讲学什么

实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。
上确界证明通常有两步： 证明它是上界； 证明任何更小的数都不是上界。

证明题通用方法（先读本页）

严格证明的四个必要动作

第一：写出假设。比如“K 非空、闭、有界”，不能只写“K 很好”。
第二：逐字展开定义。写出 $\forall$（任意）、$\exists$（存在）、$\Rightarrow$（推出）的顺序。
第三：解释每一次定理调用。从有界得到收敛子列还缺 Bolzano–Weierstrass；得到极限在原集合还需要闭性。
第四：检查结论是否恰好是题目所求。“有极限”不等于“极限属于集合”，“函数有界”不等于“达到最大值”。

最常用的三种模板

集合相等：任取 x，证明 $x \in A \Rightarrow x \in B$，再证明 $x \in B \Rightarrow x \in A$。
极限：给定任意 $\varepsilon > 0$，找出一个 N（或 δ）并验证最终不等式。
存在性：常用反证、上确界构造、嵌套区间、连续函数极值、固定点定理；对每个方法要明确其全部前提。

2 逐页课件与中文讲解

2.1 第 1 页｜本页课件

Theory of Value

An Axiomatic Analysis of Economic Equilibrium

Gerard Debreu (1959)

Professor Shunming Zhang

School of Finance
Renmin University of China

September 21, 2022

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

结合原文理解：上确界证明通常有两步： 证明它是上界； 证明任何更小的数都不是上界。

读课件时的动作：本页用于承接前后概念。阅读时要回答两个问题：这一页新增了什么条件？它将用于下一页的哪一个结论？

2.2 第 2 页 | Chapter 1: Mathematics

Chapter 1: Mathematics

01. Introduction
02. Sets
03. Functions and Correspondences
04. Preorderings
05. Real Numbers
06. Limits in $\mathbb{R}^N$
07. Continuous Functions
08. Continuous Correspondences
09. Vectors in $\mathbb{R}^N$
10. Fixed Points

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

本页重点：反身性是 xRx；传递性是 xRy 且 yRz　xRz。完全性与反对称性不包含在预序定义内。

读课件时的动作：本页用于承接前后概念。阅读时要回答两个问题：这一页新增了什么条件？它将用于下一页的哪一个结论？

2.3 第 3 页｜ Real Numbers

Real Numbers

The set $\mathbb{R}$ of (finite, real) numbers is defined as a set of elements having the following properties (all familiar, perhaps with the exception of the last one).

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

结合原文理解：上确界证明通常有两步： 证明它是上界； 证明任何更小的数都不是上界。

读课件时的动作：本页用于承接前后概念。阅读时要回答两个问题：这一页新增了什么条件？它将用于下一页的哪一个结论？

2.4 第 4 页｜ Real Numbers

Real Numbers

There are on $\mathbb{R}$ two associative and commutative binary operations (addition $+$, and multiplication $\cdot$). There is an element, 0, which, if added to any element x, gives x. Any element x has a negative, i.e., an element which, if added to x, gives 0. There is an element, 1, which if multiplied by any element x, gives x. Any element x different from 0 has an inverse, i.e., an element which, if multiplied by x, gives 1. Multiplication is distributive with respect to addition. 0 is different from 1.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

结合原文理解：上确界证明通常有两步： 证明它是上界； 证明任何更小的数都不是上界。

读课件时的动作：本页用于承接前后概念。阅读时要回答两个问题：这一页新增了什么条件？它将用于下一页的哪一个结论？

2.5 第 5 页｜ Real Numbers

Real Numbers

There is on $\mathbb{R}$ a complete ordering denoted $\leqslant$. If z is any element and $x_1 \leqslant x_2$, then $x_1 + z \leqslant x_2 + z$. If $0 \leqslant x_1$ and $0 \leqslant x_2$, then $0 \leqslant x_1 \cdot x_2$.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

结合原文理解：上确界证明通常有两步： 证明它是上界； 证明任何更小的数都不是上界。

读课件时的动作：本页用于承接前后概念。阅读时要回答两个问题：这一页新增了什么条件？它将用于下一页的哪一个结论？

2.6 第 6 页｜ Real Numbers

Real Numbers

Finally, every non-empty subset X of $\mathbb{R}$ which has an upper bound has a least upper bound, and every non-empty subset X of $\mathbb{R}$ which has a lower bound has a greatest lower bound.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

本页重点：上确界不是一定被取到的最大值。验证 s=sup A：所有 a ∈ A 满足 a≤s，并且任意 ε>0 总有 a ∈ A 满足 s−ε<a。

读课件时的动作：本页用于承接前后概念。阅读时要回答两个问题：这一页新增了什么条件？它将用于下一页的哪一个结论？

对应的完整证明：参见本讲义最后的「supremum」证明专题。

2.7 第 7 页｜ Real Numbers

Real Numbers

A few definitions and further (derived) properties of $\mathbb{R}$ will now be added. $x_1 + x_2$ is the sum of x_1 and x_2, $x_1 \cdot x_2$ is their product. If $x_1, \cdots, x_n, \cdots, x_N$ are N elements of $\mathbb{R}$, their sum is denoted $\sum_{n=1}^{N} x_n$. The product of N elements equal to x is denoted x^N and called the Nth power of x. 0 (resp. 1) is the only element having the property which defines 0 (resp. 1). Any $x \in \mathbb{R}$ has a unique negative denoted $-x$. One writes $x_1 - x_2$ for $x_1 + (-x_2)$. The corresponding binary operation is called subtraction, and the result difference. Any $x \in \mathbb{R}$ different from 0 has a unique inverse denoted $\frac{1}{x}$. One writes $\frac{x_1}{x_2}$ for $x_1 \cdot \frac{1}{x_2}$. The corresponding binary operation is called division, and the result quotient. For any $x \in \mathbb{R}$ one has $0 \cdot x = 0$, and $(-1) \cdot x = -x$. The multiplication dot will now always be dropped.

这一页怎么读

概念主线： 实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

结合原文理解： 上确界证明通常有两步： 证明它是上界； 证明任何更小的数都不是上界。

读课件时的动作： 本页以定义为主。建议将英文谓词翻译成“任意…存在…”的中文句子，再试着写一个满足定义和一个不满足定义的例子。

2.8 第 8 页｜ Real Numbers

Real Numbers

$x_1 \leqslant x_2$ (resp. $x_1 < x_2$) is read x_1 less than or equal to x_2 (resp. x_1 less than x_2). $x \leqslant 0$ (resp. $x < 0$) is read x non-positive (resp. x negative). The expressions are transposed in an obvious way if the inequality sign is inverted. One has $0 < 1$.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

结合原文理解：上确界证明通常有两步： 证明它是上界； 证明任何更小的数都不是上界。

读课件时的动作：本页用于承接前后概念。阅读时要回答两个问题：这一页新增了什么条件？它将用于下一页的哪一个结论？

2.9 第 9 页｜ Real Numbers

Real Numbers

A greatest (resp. least) element of a subset X of $\mathbb{R}$, if it exists, is unique; it is called the maximum (resp. minimum) of X and denoted $\max X$ (resp. $\min X$). One defines the absolute value $|x|$ of a number x by $|x| = \max\{x, -x\}$, and the sign of a number $x \neq 0$ by $\operatorname{sign} x = \dfrac{x}{|x|}$. One has $|x_1 + x_2| \leqslant |x_1| + |x_2|$.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

结合原文理解：上确界证明通常有两步： 证明它是上界； 证明任何更小的数都不是上界。

读课件时的动作：本页以定义为主。建议将英文谓词翻译成“任意…存在…”的中文句子，再试着写一个满足定义和一个不满足定义的例子。

2.10 第 10 页｜ Real Numbers

Real Numbers

A least upper bound of a subset X of $\mathbb{R}$, if it exists, is unique (1.4.f); it is called the supremum of X and denoted $\sup X$. Every non-empty subset X of $\mathbb{R}$ which has a lower bound has a unique greatest lower bound called the infimum of X and denoted $\inf X$.

The simplest example is provided by the "non-empty interval $I = [a, b)$. It has no maximum, its supremum is b. Its minimum, and therefore (1.4.g) its infimum, is a. See also example (2) of 1.5.o.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

本页重点：上确界不是一定被取到的最大值。验证 s=sup A：所有 a ∈ A 满足 a≤s，并且任意 ε>0 总有 a ∈ A 满足 s−ε<a。

读课件时的动作：本页有例子或构造。自己换一个简单数值例子，算出具体集合或函数值；再思考一旦撤掉某条假设，结论是否还成立。

对应的完整证明：参见本讲义最后的「supremum」证明专题。

2.11 第 11 页｜ Real Numbers

Real Numbers

The infimum (resp. supremum) of a real interval is also called its origin (resp. extremity). The length of an interval with origin a and extremity b is $b - a$.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

本页重点：上确界不是一定被取到的最大值。验证 s=sup A：所有 a ∈ A 满足 a≤s，并且任意 ε>0 总有 a ∈ A 满足 s−ε<a。

读课件时的动作：本页以定义为主。建议将英文谓词翻译成“任意…存在…”的中文句子，再试着写一个满足定义和一个不满足定义的例子。

对应的完整证明：参见本讲义最后的「supremum」证明专题。

2.12 第 12 页｜ Real Numbers

Real Numbers

By repeated addition of 1 to 0, and repeated subtraction of 1 from 0, one obtains the set $\mathbb{Z}$ of integers (non-negative and non-positive) as a subset of $\mathbb{R}$.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

结合原文理解：上确界证明通常有两步： 证明它是上界； 证明任何更小的数都不是上界。

读课件时的动作：本页用于承接前后概念。阅读时要回答两个问题：这一页新增了什么条件？它将用于下一页的哪一个结论？

2.13 第 13 页｜ Real Numbers

Real Numbers

A real number of the form $\frac{p}{q}$, where $p \in \mathbb{Z}$ and $q \in \mathbb{Z}$, and $q \neq 0$, is called a rational number. The set of rationals, a subset of $\mathbb{R}$, is denoted by $\mathbb{Q}$.

Theorem 1. The set $\mathbb{Q}$ is countable.

(use (2) of 1.3.n to prove that $\mathbb{Z}$, then $\mathbb{Q}$ is countable) and satisfies all the axioms in the definition of $\mathbb{R}$ with the exception of the last one. (For example, one can prove that the set $\{x \in \mathbb{Q} \mid x^2 < 2\}$ has no least upper bound in $\mathbb{Q}$, using the fact that there is no rational y satisfying $y^2 = 2$.)

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

本页重点：上确界不是一定被取到的最大值。验证 s=sup A：所有 a ∈ A 满足 a≤s，并且任意 ε>0 总有 a ∈ A 满足 s−ε<a。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

对应的完整证明：参见本讲义最后的「supremum」证明专题。

2.14 第 14 页｜ Real Numbers

Real Numbers

One can prove that the set $\mathbb{R}$ is not countable. One can also prove that

Theorem 2. If x, y belong to $\mathbb{R}$ and satisfy $x < y$, there is a rational r such that $x < r < y$.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

结合原文理解：上确界证明通常有两步： 证明它是上界； 证明任何更小的数都不是上界。

读课件时的动作：本页是定理或命题。先明确量词、集合范围、连续性及闭/紧/凸假设；再检查结论是“存在”“唯一”还是“对于任意”。定理本身不能替代证明。

2.15 第 15 页｜ Real Numbers

Real Numbers

Given a number $x \geqslant 0$ and a positive integer N, one can prove that there is a unique $y \geqslant 0$ such that $y^N = x$. One calls y the Nth root of x, and one writes $y = x^{\frac{1}{N}}$.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

结合原文理解：上确界证明通常有两步： 证明它是上界； 证明任何更小的数都不是上界。

读课件时的动作：本页用于承接前后概念。阅读时要回答两个问题：这一页新增了什么条件？它将用于下一页的哪一个结论？

2.16 第 16 页｜ Real Numbers

Real Numbers

Consider a sequence (x^q) of numbers. Intuitively, one says that (x^q) converges (or tends) to the number x^0 if x^q is as close to x^0 as one wishes provided that q is large enough. In a precise fashion, (x^q) converges to x^0 if, for any number $\varepsilon > 0$, there is an integer $q' > 0$ (depending on ε) such that $q > q'$ implies $\left|x^q - x^0\right| < \varepsilon$. One writes $x^q \to x^0$.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

结合原文理解：上确界证明通常有两步： 证明它是上界； 证明任何更小的数都不是上界。

读课件时的动作：本页有例子或构造。自己换一个简单数值例子，算出具体集合或函数值；再思考一旦撤掉某条假设，结论是否还成立。

2.17 第 17 页｜ Real Numbers

Real Numbers

A sequence which tends to a number is called convergent. It is clear that "$x^q \to x^0$ and $x^q \to x^*$" implies "$x^0 = x^*$"; thus a convergent sequence tends to a unique number called its limit.

Example: (1) The sequence $\left(\frac{(-1)^q}{q}\right)$ tends to 0 since, given $\varepsilon > 0$, making $\frac{1}{q} < \varepsilon$ is equivalent to making $q > \frac{1}{\varepsilon}$. (2) Note that if X is the set $\left\{\frac{1}{1}, \frac{1}{2}, \cdots, \frac{1}{q}, \cdots\right\}$, then X has no minimum and $\inf X = 0$. (3) The sequences $((-1)^q)$ and (q) are not convergent.

这一页怎么读

概念主线： 实数的核心是有序域加完备性。最小上界不是简单的"最大元素"；上确界可以不属于原集合。

结合原文理解： 上确界证明通常有两步： 证明它是上界； 证明任何更小的数都不是上界。

读课件时的动作： 本页出现推导或证明。把每一行改写为"由哪条定义或假设，得到哪一个新结论"。若老师写了"显然"，不要跳过：你至少应能写出一条完整的蕴含式。

2.18 第 18 页｜ Real Numbers

Real Numbers

The elements of $\mathbb{R}$ are also called points. The set $\mathbb{R}$ may be visualized as follows. On a horizontal straight line, two different points are chosen; they will represent 0 and 1. 1 being to the right of 0. An element x of $\mathbb{R}$ is then represented by a point of the straight line at distance $|x|$ from 0, to the right (resp. left) of 0 if x is positive (resp. negative).

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

结合原文理解：上确界证明通常有两步： 证明它是上界； 证明任何更小的数都不是上界。

读课件时的动作：本页以定义为主。建议将英文谓词翻译成“任意…存在…”的中文句子，再试着写一个满足定义和一个不满足定义的例子。

2.19 第 19 页｜ Real Numbers

Real Numbers

The letters $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$ have throughout this volume the meanings introduced respectively in 1.2.a, 1.5.j, 1.5.k, 1.5.a.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

结合原文理解：上确界证明通常有两步： 证明它是上界； 证明任何更小的数都不是上界。

读课件时的动作：本页用于承接前后概念。阅读时要回答两个问题：这一页新增了什么条件？它将用于下一页的哪一个结论？

2.20 第 20 页｜ Real Number System

Real Number System

There are at least two approaches to defining the real number system.

Approach 1. Start with the set of natural numbers $\mathbb{N}$, construct the set $\mathbb{Z}$ of all integers $(\mathbb{N} \cup \{0\} \cup (-\mathbb{N}))$, and next, the set $\mathbb{Q}$ of rationals and then the set of $\mathbb{R}$ of real numbers either as the set of all Cauchy sequences of rationals or as Dedekind cuts. The step going from $\mathbb{Q}$ to $\mathbb{R}$ via Cauchy sequences is also available for completing any incomplete metric space.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

本页重点：有理数缺少某些极限点，例如逼近 $\sqrt{2}$ 的有理数数列。构造实数就是要填补这样的缺口。

读课件时的动作：本页用于承接前后概念。阅读时要回答两个问题：这一页新增了什么条件？它将用于下一页的哪一个结论？

对应的完整证明：参见本讲义最后的「complete」证明专题。

2.21 第 21 页｜Real Number System

Real Number System

There are at least two approaches to defining the real number system.

Approach 2. Define the set of real numbers, $\mathbb{R}$, as a set that satisfies three sets of axioms. The first set is algebraic involving addition and multiplication. The second set is on ordering that, with the first, makes $\mathbb{R}$ an ordered field (Royden 1988). The third set is a single axiom known as the completeness axiom. Thus $\mathbb{R}$ is defined as complete ordered field.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

结合原文理解：上确界证明通常有两步： 证明它是上界； 证明任何更小的数都不是上界。

读课件时的动作：本页以定义为主。建议将英文谓词翻译成“任意…存在…”的中文句子，再试着写一个满足定义和一个不满足定义的例子。

2.22 第 22 页 | Additive Group

Additive Group

A group is a set, G, together with an operation $+$ (called the group law of G) that combines any two elements a_1 and a_2 to form another element, denoted $a_1 + a_2$. To qualify as a group, the set and operation, $(G, +)$, must satisfy four requirements known as the group axioms:

(1) Closure: For all a_1 and a_2 in G, the result of the operation, $a_1 + a_2$, is also in G.

(2) Associativity: For all a_1, a_2 and a_3 in G, $(a_1 + a_2) + a_3 = a_1 + (a_2 + a_3)$.

(3) Identity element: There exists an element 0 in G such that, for every element a in G, the equation $0 + a = a + 0 = a$ holds. Such an element is unique, and thus one speaks of the identity element.

(4) Additive Inverse: For each a in G, there exists an inverse element $-a$ in G such that $a + (-a) = (-a) + a = 0$.

A group $(G, +)$ with the property that $a_1 + a_2 = a_2 + a_1$ for all a_1 and a_2 in G is called Abelian (in honor of Niels Henrik Abel) or commutative. Groups not satisfying this property are said to be non-Abelian or non-commutative.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

结合原文理解：上确界证明通常有两步： 证明它是上界； 证明任何更小的数都不是上界。

读课件时的动作：本页以定义为主。建议将英文谓词翻译成“任意…存在…”的中文句子，再试着写一个满足定义和一个不满足定义的例子。

2.23 第 23 页｜Multiplicative Group

Multiplicative Group

A group is a set, G, together with an operation $\times$ (called the group law of G) that combines any two elements a_1 and a_2 to form another element, denoted $a_1 \times a_2$. To qualify as a group, the set and operation, $(G, \times)$, must satisfy four requirements known as the group axioms:

(1) Closure: For all a_1 and a_2 in G, the result of the operation, $a_1 \times a_2$, is also in G.

(2) Associativity: For all a_1, a_2 and a_3 in G, $(a_1 \times a_2) \times a_3 = a_1 \times (a_2 \times a_3)$.

(3) Identity element: There exists an element 1 in G such that, for every element a in G, the equation $1 \times a = a \times 1 = a$ holds. Such an element is unique, and thus one speaks of the identity element.

(4) Multiplicative Inverse: For each a in G, there exists an inverse element a^{-1} in G such that $a \times a^{-1} = a^{-1} \times a = 1$.

A group $(G, \times)$ with the property that $a_1 \times a_2 = a_2 \times a_1$ for all a_1 and a_2 in G is called Abelian (in honor of Niels Henrik Abel) or commutative. Groups not satisfying this property are said to be non-Abelian or non-commutative.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

结合原文理解：上确界证明通常有两步： 证明它是上界； 证明任何更小的数都不是上界。

读课件时的动作：本页以定义为主。建议将英文谓词翻译成“任意…存在…”的中文句子，再试着写一个满足定义和一个不满足定义的例子。

2.24 第 24 页 | Group

Group

A group is a set, G, together with an operation $\cdot$ (called the group law of G) that combines any two elements a_1 and a_2 to form another element, denoted $a_1 \cdot a_2$. To qualify as a group, the set and operation, $(G, \cdot)$, must satisfy four requirements known as the group axioms:

(1) Closure: For all a_1 and a_2 in G, the result of the operation, $a_1 \cdot a_2$, is also in G.

(2) Associativity: For all a_1, a_2 and a_3 in G, $(a_1 \cdot a_2) \cdot a_3 = a_1 \cdot (a_2 \cdot a_3)$.

(3) Identity element: There exists an element e in G such that, for every element a in G, the equation $e \cdot a = a \cdot e = a$ holds. Such an element is unique, and thus one speaks of the identity element.

(4) Inverse element: For each a in G, there exists an inverse element a^{-1} in G such that $a \cdot a^{-1} = a^{-1} \cdot a = e$.

A group $(G, \cdot)$ with the property that $a_1 \cdot a_2 = a_2 \cdot a_1$ for all a_1 and a_2 in G is called Abelian (in honor of Niels Henrik Abel) or commutative. Groups not satisfying this property are said to be non-Abelian or non-commutative.

这一页怎么读

概念主线： 实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

结合原文理解： 上确界证明通常有两步： 证明它是上界； 证明任何更小的数都不是上界。

读课件时的动作： 本页以定义为主。建议将英文谓词翻译成“任意…存在…”的中文句子，再试着写一个满足定义和一个不满足定义的例子。

2.25 第 25 页｜ Ring

Ring

A ring is a set R equipped with two binary operations $+$ and $\cdot$ satisfying the following three sets of axioms, called the ring axioms.

(1) R is an abelian group under addition $+$, meaning that:

(1.1) Commutativity: For all a_1 and a_2 in R, the result of the operation, $a_1 + a_2 = a_2 + a_1$, is also in R.

(1.2) Associativity: For all a_1, a_2 and a_3 in R, $(a_1 + a_2) + a_3 = a_1 + (a_2 + a_3)$.

(1.3) Additive identity: There exists an element 0 in R such that, for every element a in R, the equation $0 + a = a + 0 = a$ holds.

(1.4) Additive Inverse: For each a in R, there exists an inverse element $-a$ in R such that $a + (-a) = (-a) + a = 0$.

(2) R is a monoid under multiplication $\times$, meaning that:

(2.1) Closure: For all a_1 and a_2 in R, the result of the operation, $a_1 \times a_2$, is also in R.

(2.2) Associativity: For all a_1, a_2 and a_3 in R, $(a_1 \times a_2) \times a_3 = a_1 \times (a_2 \times a_3)$.

(2.3) Multiplicative Identity: There exists an element 1 in R such that, for every element a in R, the equation $1 \times a = a \times 1 = a$ holds.

(3) Multiplication is distributive with respect to addition:

(3.1) Left Distributivity: For all a_1, a_2 and a_3 in R, $a_1 \times (a_2 + a_3) = a_1 \times a_2 + a_1 \times a_3$.

(3.2) Right Distributivity: For all a_1, a_2 and a_3 in R, $(a_2 + a_3) \times a_1 = a_2 \times a_1 + a_3 \times a_1$.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

结合原文理解：上确界证明通常有两步： 证明它是上界； 证明任何更小的数都不是上界。

读课件时的动作：本页以定义为主。建议将英文谓词翻译成“任意…存在…”的中文句子，再试着写一个满足定义和一个不满足定义的例子。

2.26 第 26 页｜本页课件

Abelian Ring = Commutative Ring

A ring is a set R equipped with two binary operations $+$ and $\cdot$ satisfying the following three sets of axioms, called the ring axioms.

(1) R is an abelian group under addition $+$, meaning that:

(1.1) Commutativity: For all a_1 and a_2 in R, the result of the operation, $a_1 + a_2 = a_2 + a_1$, is also in R.

(1.2) Associativity: For all a_1, a_2 and a_3 in R, $(a_1 + a_2) + a_3 = a_1 + (a_2 + a_3)$.

(1.3) Additive identity: There exists an element 0 in R such that, for every element a in R, the equation $0 + a = a + 0 = a$ holds.

(1.4) Additive Inverse: For each a in R, there exists an inverse element $-a$ in R such that $a + (-a) = (-a) + a = 0$.

(2) R is a monoid under multiplication $\times$, meaning that:

(2.1) Commutativity: For all a_1 and a_2 in R, the result of the operation, $a_1 \times a_2 = a_2 \times a_1$, is also in R.

(2.2) Associativity: For all a_1, a_2 and a_3 in R, $(a_1 \times a_2) \times a_3 = a_1 \times (a_2 \times a_3)$.

(2.3) Multiplicative Identity: There exists an element 1 in R such that, for every element a in R, the equation $1 \times a = a \times 1 = a$ holds.

(3) Multiplication is distributive with respect to addition:

(3.1) Left Distributivity: For all a_1, a_2 and a_3 in R, $a_1 \times (a_2 + a_3) = a_1 \times a_2 + a_1 \times a_3$.

(3.2) Right Distributivity: For all a_1, a_2 and a_3 in R, $(a_2 + a_3) \times a_1 = a_2 \times a_1 + a_3 \times a_1$.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

结合原文理解：上确界证明通常有两步： 证明它是上界； 证明任何更小的数都不是上界。

读课件时的动作：本页用于承接前后概念。阅读时要回答两个问题：这一页新增了什么条件？它将用于下一页的哪一个结论？

2.27 第 27 页｜本页课件

Field

A field is a set F equipped with two binary operations $+$ and $\cdot$ satisfying the following three sets of axioms, called the ring axioms.

(1) F is an abelian group under addition $+$, meaning that:

(1.1) Commutativity: For all a_1 and a_2 in F, the result of the operation, $a_1 + a_2 = a_2 + a_1$, is also in F.

(1.2) Associativity: For all a_1, a_2 and a_3 in F, $(a_1 + a_2) + a_3 = a_1 + (a_2 + a_3)$.

(1.3) Additive identity: There exists an element 0 in F such that, for every element a in F, the equation $0 + a = a + 0 = a$ holds.

(1.4) Additive Inverse: For each a in F, there exists an inverse element $-a$ in F such that $a + (-a) = (-a) + a = 0$.

(2) F is a monoid under multiplication $\times$, meaning that:

(2.1) Commutativity: For all a_1 and a_2 in F, the result of the operation, $a_1 \times a_2 = a_2 \times a_1$, is also in F.

(2.2) Associativity: For all a_1, a_2 and a_3 in F, $(a_1 \times a_2) \times a_3 = a_1 \times (a_2 \times a_3)$.

(2.3) Multiplicative Identity: There exists an element 1 in F such that, for every element a in F, the equation $1 \times a = a \times 1 = a$ holds.

(2.4) Multiplicative Inverse: For each $a \neq 0$ in F, there exists an inverse element a^{-1} in F such that $a \times a^{-1} = a^{-1} \times a = 1$.

(3) Multiplication is distributive with respect to addition:

(3.1) Left Distributivity: For all a_1, a_2 and a_3 in F, $a_1 \times (a_2 + a_3) = a_1 \times a_2 + a_1 \times a_3$.

(3.2) Right Distributivity: For all a_1, a_2 and a_3 in R, $(a_2 + a_3) \times a_1 = a_2 \times a_1 + a_3 \times a_1$.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

结合原文理解：上确界证明通常有两步： 证明它是上界； 证明任何更小的数都不是上界。

读课件时的动作：本页用于承接前后概念。阅读时要回答两个问题：这一页新增了什么条件？它将用于下一页的哪一个结论？

2.28 第 28 页｜ Field

Field

If there is an element 1 in R such that $1 \neq 0$ and $1 \times a = a \times 1 = a$ for each element a in R, we say that R is a ring with unity or identity. A commutative ring R with identity is called an integral domain if, for every a_1 and a_2 in R such that $a_1 \times a_2 = 0$, either $a_1 = 0$ or $a_2 = 0$. A division ring is a ring R, with an identity, in which every non-zero element in R is a unit; that is, for each a in R with $a \neq 0$, there exists a unique element a^{-1} such that $a^{-1} \times a = a \times a^{-1} = 1$. A commutative division ring is called a field.

Rings: $\mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$

Fields: $\mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

结合原文理解：上确界证明通常有两步： 证明它是上界； 证明任何更小的数都不是上界。

读课件时的动作：本页以定义为主。建议将英文谓词翻译成“任意…存在…”的中文句子，再试着写一个满足定义和一个不满足定义的例子。

2.29 第 29 页｜Linear Space = Vector Space

Linear Space = Vector Space

A vector space over a field F is a set V together with two operations that satisfy the eight axioms listed below.

(1) The first operation, called vector addition or simply addition $+ : V \times V \to V$, takes any two vectors v_1 and v_2 and assigns to them a third vector which is commonly written as $v_1 + v_2$, and called the sum of these two vectors. (Note that the resultant vector is also an element of the set V).

(2) The second operation, called scalar multiplication $\cdot : F \times V \to V$, takes any scalar a and any vector v and gives another vector $a \cdot v$. (Similarly, the vector $a \cdot v$ is an element of the set V).

Elements of V are commonly called vectors. Elements of F are commonly called scalars.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

结合原文理解：上确界证明通常有两步： 证明它是上界； 证明任何更小的数都不是上界。

读课件时的动作：本页以定义为主。建议将英文谓词翻译成“任意…存在…”的中文句子，再试着写一个满足定义和一个不满足定义的例子。

2.30 第 30 页｜Linear Space = Vector Space

Linear Space = Vector Space

(1) Commutativity: For all v_1 and v_2 in F, the result of the operation, $v_1 + v_2 = v_2 + v_1$, is also in V.

(2) Associativity: For all v_1, v_2 and v_3 in V, $(v_1 + v_2) + v_3 = v_1 + (v_2 + v_3)$.

(3) Additive identity: There exists an element 0 in V such that, for every element v in V, the equation $0 + v = v + 0 = v$ holds.

(4) Additive Inverse: For each v in V, there exists an inverse element $-v$ in V such that $v + (-v) = (-v) + v = 0$.

(5) Multiplicative Identity in Field F for Scalar Multiplication: For each v in V, $1 \cdot v = v$.

(6) Compatibility of Scalar Multiplication with Field Multiplication: For any a_1 and a_2 in F and each v in V, $a_1 \cdot (a_2 \cdot v) = (a_1 \times a_2) \cdot v$.

(7) Distributivity of Scalar Multiplication with respect to Vector Addition: For any a in F and each v_1 and v_2 in V, $a \cdot (v_1 + v_2) = a \cdot v_1 + a \cdot v_2$.

(8) Distributivity of Scalar Multiplication with respect to Field Addition: For any a_1 and a_2 in F and each v in V, $(a_1 + a_2) \cdot v = a_1 \cdot v + a_2 \cdot v$.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

结合原文理解：上确界证明通常有两步： 证明它是上界； 证明任何更小的数都不是上界。

读课件时的动作：本页用于承接前后概念。阅读时要回答两个问题：这一页新增了什么条件？它将用于下一页的哪一个结论？

2.31 第 31 页｜Order Axioms

Order Axioms

Order Axioms There is a subset $\mathbb{P}$ of the set $\mathbb{R}$ of real numbers, called the set of strictly positive numbers such that:

(1) For any real number a in $\mathbb{R}$, exactly one of the following holds: $a = 0$ or $a \in \mathbb{P}$ or $-a \in \mathbb{P}$.

(2) If a_1 and a_2 in $\mathbb{P}$, $a_1 + a_2 \in \mathbb{P}$.

(3) If a_1 and a_2 in $\mathbb{P}$, $a_1 \cdot a_2 \in \mathbb{P}$.

The "Less Than" Relation We now define

- $a_1 < a_2$ to mean $a_2 - a_1 \in \mathbb{P}$ $(a_2 - a_1 \in \mathbb{R}_{++})$
- $a_1 \leqslant a_2$ to mean $a_2 - a_1 \in \mathbb{P}$ or $a_1 = a_2$ $(a_2 - a_1 \in \mathbb{R}_{+})$
- $a_1 > a_2$ to mean $a_1 - a_2 \in \mathbb{P}$ $(a_1 - a_2 \in \mathbb{R}_{++})$
- $a_1 \geqslant a_2$ to mean $a_1 - a_2 \in \mathbb{P}$ or $a_1 = a_2$ $(a_1 - a_2 \in \mathbb{R}_{+})$

$a_1 < a_2$ if and only if $a_2 > a_1$ and $a_1 \leqslant a_2$ if and only if $a_2 \geqslant a_1$.

$$\mathbb{P} = \mathbb{R}_{+} \setminus \{0\} = \mathbb{R}_{++}$$

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

结合原文理解：上确界证明通常有两步： 证明它是上界； 证明任何更小的数都不是上界。

读课件时的动作：本页以定义为主。建议将英文谓词翻译成“任意…存在…”的中文句子，再试着写一个满足定义和一个不满足定义的例子。

2.32 第 32 页｜ Ordered Field

Ordered Field

Any field F, together with two operations $\oplus$ and $\otimes$ and two members $0_{\oplus}$ and $0_{\otimes}$ of F, and a subset P of F, which satisfies the corresponding versions of (1) - (2) - (3), is called an ordered field.

Example. Both $\mathbb{Q}$ and $\mathbb{R}$ are ordered fields.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

结合原文理解：上确界证明通常有两步： 证明它是上界； 证明任何更小的数都不是上界。

读课件时的动作：本页有例子或构造。自己换一个简单数值例子，算出具体集合或函数值；再思考一旦撤掉某条假设，结论是否还成立。

2.33 第 33 页｜本页课件

Completeness Axiom

Dedekind Completeness Axiom Suppose A and B are two (non-empty) sets of real numbers with the properties:

(1) if $a \in A$ and $b \in B$, then $a < b$.

(2) every real number is in either A or B (in symbols; $A \cup B = \mathbb{R}$).

Then there is a unique real number c such that:

(1) if $a < c$ then $a \in A$, and

(2) if $b > c$ then $b \in B$.

Note that every number strictly less than c belongs to A and every number strictly more than c belongs to B. Moreover, either $c \in A$ or $c \in B$ by (2). Hence if $c \in A$ then $A = (-\infty, c]$ and $B = (c, \infty)$; while if $c \in B$ then $A = (-\infty, c)$ and $B = [c, \infty)$.

The pair of sets $\{A, B\}$ is called a Dedekind Cut.

The intuitive idea of Dedekind Completeness Axiom is that there are no "holes" in the real numbers.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

本页重点：有理数缺少某些极限点，例如逼近 $\sqrt{2}$ 的有理数数列。构造实数就是要填补这样的缺口。

读课件时的动作：本页以定义为主。建议将英文谓词翻译成“任意…存在…”的中文句子，再试着写一个满足定义和一个不满足定义的例子。

对应的完整证明：参见本讲义最后的「complete」证明专题。

2.34 第 34 页｜ Dedekind - Cantor Axiom of Continuity of a Real Line

Dedekind - Cantor Axiom of Continuity of a Real Line

Dedekind - Cantor Axiom of Continuity of a Real Line Suppose the pair of sets (A, A^c) is a Dedekind Cut of $\mathbb{R}$ with the property: if $a \in A$ and $b \in A^c$, then $a < b$. Then either there exists a maximum in A or there exists a minimum in A^c:

(1) If there exists a maximum c in A, then $A = (-\infty, c]$ and $A^c = (c, \infty)$.

(2) If there exists a minimum c in A^c, then $A = (-\infty, c)$ and $A^c = [c, \infty)$.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

本页重点：有理数缺少某些极限点，例如逼近 $\sqrt{2}$ 的有理数数列。构造实数就是要填补这样的缺口。

读课件时的动作：本页有例子或构造。自己换一个简单数值例子，算出具体集合或函数值；再思考一旦撤掉某条假设，结论是否还成立。

对应的完整证明：参见本讲义最后的「complete」证明专题。

2.35 第 35 页｜ Supremum and Infimum

Supremum and Infimum

(1) Let S be a set of real numbers. u is an upper bound for S if $x \leqslant u$ for all $x \in S$. A set S is bounded above if it has an upper bound.

(2) b is the least upper bound (or l.u.b. or supremum or sup) for S if b is an upper bound, and moreover $b \leqslant u$ whenever u is any upper bound for S.

---

(1) Let S be a set of real numbers. d is an lower bound for S if $x \geqslant d$ for all $x \in S$. A set S is bounded below if it has a lower bound.

(2) l is the greatest lower bound (or g.l.b. or infimum or inf) for S if l is an lower bound, and moreover $l \geqslant d$ whenever d is any lower bound for S.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

本页重点：上确界不是一定被取到的最大值。验证 s=sup A：所有 a ∈ A 满足 a≤s，并且任意 ε>0 总有 a ∈ A 满足 s−ε<a。

读课件时的动作：本页用于承接前后概念。阅读时要回答两个问题：这一页新增了什么条件？它将用于下一页的哪一个结论？

对应的完整证明：参见本讲义最后的「supremum」证明专题。

2.36 第 36 页｜Supremum and Infimum Completeness Axiom

Supremum and Infimum Completeness Axiom

Theorem of Supremum (Least Upper Bound Completeness Axiom)
Suppose S is a non-empty set of real numbers which is bounded above. Then S has a supremum in $\mathbb{R}$.

Theorem of Infimum (Greatest Lower Bound Completeness Axiom)
Suppose S is a non-empty set of real numbers which is bounded below. Then S has a infimum in $\mathbb{R}$.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

本页重点：上确界不是一定被取到的最大值。验证 s=sup A：所有 a A 满足 a≤s，并且任意 ε>0 总有 a A 满足 s−ε<a。

读课件时的动作：本页是定理或命题。先明确量词、集合范围、连续性及闭/紧/凸假设；再检查结论是“存在”“唯一”还是“对于任意”。定理本身不能替代证明。

对应的完整证明：参见本讲义最后的「supremum」证明专题。

2.37 第 37 页｜本页课件

Dedekind - Cantor Axiom vs Supremum and Infimum Completeness Axiom

[Necassity] Suppose that S is a non-empty set of real numbers which is bounded above. Let

$$B = \{x \in \mathbb{R} : x \text{ is an upper bound for } S\} \quad \text{and} \quad A = \mathbb{R} \setminus B.$$

Note that $B \neq \emptyset$; and if $x \in S$ then $x - 1$ is not an upper bound for S so $A \neq \emptyset$.

Suppose $a \in A$ and $b \in B$, then $a < b$. If $a \geqslant b$, then a would also be an upper bound for S, which contradicts the definition of A.

Let c be the real number given by Dedekind - Cantor Axiom. We claim that $c = \sup S$.

If $c \in A$ then c is not an upper bound for S and so there exists $x \in S$ with $c < x$. But then $a = \dfrac{c + x}{2}$ is not an upper bound for S, i.e. $a \in A$, contradicting the fact from the conclusion of Dedekind - Cantor Axiom that $a \leqslant c$ for all $a \in A$. Hence $c \in B$. But if $c \in B$ then $c \leqslant b$ for all $b \in B$; i.e. c is less than any upper bound for S.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

本页重点：有理数缺少某些极限点，例如逼近 $\sqrt{2}$ 的有理数数列。构造实数就是要填补这样的缺口。

读课件时的动作：本页有例子或构造。自己换一个简单数值例子，算出具体集合或函数值；再思考一旦撤掉某条假设，结论是否还成立。

对应的完整证明：参见本讲义最后的「complete」证明专题。

2.38 第 38 页｜ Dedekind - Cantor Axiom vsSupremum and Infimum Complete

Dedekind - Cantor Axiom vs Supremum and Infimum Completeness Axiom

[Sufficiency] For this, suppose $\{A, B\}$ is a Dedekind cut. Then A is bounded above (by any element of B). Let $c = \sup A$. We claim that

$$a < c \Rightarrow a \in A, \qquad \text{and} \qquad b > c \Rightarrow b \in B.$$

- Suppose $a < c$. Now every member of B is an upper bound for A, from the first property of a Dedekind cut; hence $a \notin B$, as otherwise a would be an upper bound for A which is less than the least upper bound c. Hence $a \in A$.
- Next suppose $b > c$. Since c is an upper bound for A (in fact the least upper bound), it follows we cannot have $b \in A$, and thus $b \in B$.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

本页重点：上确界不是一定被取到的最大值。验证 s=sup A：所有 a ∈ A 满足 a≤s，并且任意 ε>0 总有 a ∈ A 满足 s−ε<a。

读课件时的动作：本页有例子或构造。自己换一个简单数值例子，算出具体集合或函数值；再思考一旦撤掉某条假设，结论是否还成立。

对应的完整证明：参见本讲义最后的「supremum」证明专题。

2.39 第 39 页｜本页课件

Dedekind - Cantor Axiom vs Supremum and Infimum Completeness Axiom

[Necassity] Suppose that S is a non-empty set of real numbers which is bounded below. Let

$$A = \{x \in \mathbb{R} : x \text{ is a lower bound for } S\} \quad \text{and} \quad B = \mathbb{R} \setminus A.$$

Note that $A \neq \emptyset$; and if $x \in S$ then $x + 1$ is not a lower bound for S so $B \neq \emptyset$.

Suppose $a \in A$ and $b \in B$, then $a < b$. If $a \geqslant b$, then b would also be a lower bound for S, which contradicts the definition of B.

Let c be the real number given by Dedekind - Cantor Axiom. We claim that $c = \inf S$.

If $c \in B$ then c is not a lower bound for S and so there exists $x \in S$ with $x < c$. But then $b = \dfrac{x + c}{2}$ is not a lower bound for S, i.e. $b \in B$, contradicting the fact from the conclusion of Dedekind - Cantor Axiom that $c \leqslant b$ for all $b \in B$. Hence $c \in A$. But if $c \in A$ then $a \leqslant c$ for all $a \in A$; i.e. c is more than any lower bound for S.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

本页重点：有理数缺少某些极限点，例如逼近 $\sqrt{2}$ 的有理数数列。构造实数就是要填补这样的缺口。

读课件时的动作：本页有例子或构造。自己换一个简单数值例子，算出具体集合或函数值；再思考一旦撤掉某条假设，结论是否还成立。

对应的完整证明：参见本讲义最后的「complete」证明专题。

2.40 第 40 页｜ Dedekind - Cantor Axiom vsSupremum and Infimum Complete

Dedekind - Cantor Axiom vs Supremum and Infimum Completeness Axiom

[Sufficiency] For this, suppose $\{A, B\}$ is a Dedekind cut. Then B is bounded below (by any element of A). Let $c = \inf B$. We claim that

$$a < c \Rightarrow a \in A, \qquad \text{and} \qquad b > c \Rightarrow b \in B.$$

- Suppose $a < c$. Since c is a lower bound for B (in fact the greatest lower bound), it follows we cannot have $a \in B$, and thus $a \in A$.
- Next suppose $c < b$. Now every member of A is a lower bound for B, from the first property of a Dedekind cut; hence $b \notin A$, as otherwise b would be a lower bound for B which is less than the greatest lower bound c. Hence $b \in B$.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

本页重点：上确界不是一定被取到的最大值。验证 s=sup A：所有 a ∈ A 满足 a≤s，并且任意 ε>0 总有 a ∈ A 满足 s−ε<a。

读课件时的动作：本页有例子或构造。自己换一个简单数值例子，算出具体集合或函数值；再思考一旦撤掉某条假设，结论是否还成立。

对应的完整证明：参见本讲义最后的「supremum」证明专题。

2.41 第 41 页｜Proposition on Supremum and Infimum

Proposition on Supremum and Infimum

Suppose S is a non-empty set of real numbers. Then $l = \inf S$ iff

(1) $x \geqslant l$ for all $x \in S$;

(2) for each $\varepsilon > 0$ there exists $x \in S$ such that $x < l + \varepsilon$.

Suppose S is a non-empty set of real numbers. Then $b = \sup S$ iff

(1) $x \leqslant b$ for all $x \in S$;

(2) for each $\varepsilon > 0$ there exists $x \in S$ such that $x > b - \varepsilon$.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

本页重点：上确界不是一定被取到的最大值。验证 s=sup A：所有 a A 满足 a≤s，并且任意 ε>0 总有 a A 满足 s−ε<a。

读课件时的动作：本页有例子或构造。自己换一个简单数值例子，算出具体集合或函数值；再思考一旦撤掉某条假设，结论是否还成立。

对应的完整证明：参见本讲义最后的「supremum」证明专题。

2.42 第 42 页｜Definition of Supremum

Definition of Supremum

Suppose S is a non-empty set of real numbers. Then $b = \sup S$ iff

(1) $x \leqslant b$ for all $x \in S$;

(2) $b \leqslant u$ whenever u is any upper bound for S.

Suppose S is a non-empty set of real numbers. Then $b = \sup S$ iff

(1) $x \leqslant b$ for all $x \in S$;

(2′) for each $\varepsilon > 0$ there exists $x \in S$ such that $x > b - \varepsilon$.

$[(1) + (2) \Rightarrow (2')]$ Suppose $(2')$ is not true. Then for some $\varepsilon > 0$ it follows that $x \leqslant b - \varepsilon$ for every $x \in S$, i.e. $b - \varepsilon$ is an upper bound for S. This is a contradiction.

$[(1) + (2') \Rightarrow (2)]$ If $u < b$ then from $(2')$ it follows that u is not an upper bound of S. That is (2): $b \leqslant u$ whenever u is any upper bound for S.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

本页重点：上确界不是一定被取到的最大值。验证 s=sup A：所有 a ∈ A 满足 a≤s，并且任意 ε>0 总有 a ∈ A 满足 s−ε<a。

读课件时的动作：本页有例子或构造。自己换一个简单数值例子，算出具体集合或函数值；再思考一旦撤掉某条假设，结论是否还成立。

对应的完整证明：参见本讲义最后的「supremum」证明专题。

2.43 第 43 页｜Definition of Infimum

Definition of Infimum

Suppose S is a non-empty set of real numbers. Then $l = \inf S$ iff

(1) $x \geqslant l$ for all $x \in S$;

(2) $l \geqslant d$ whenever d is any lower bound for S.

Suppose S is a non-empty set of real numbers. Then $l = \inf S$ iff

(1) $x \geqslant l$ for all $x \in S$;

(2′) for each $\varepsilon > 0$ there exists $x \in S$ such that $x < l + \varepsilon$.

$[(1) + (2) \Rightarrow (2')]$ Suppose $(2')$ is not true. Then for some $\varepsilon > 0$ it follows that $x \geqslant l + \varepsilon$ for every $x \in S$, i.e. $l + \varepsilon$ is a lower bound for S. This is a contradiction.

$[(1) + (2') \Rightarrow (2)]$ If $d > l$ then from $(2')$ it follows that d is not a lower bound of S. That is (2): $l \geqslant d$ whenever d is any lower bound for S.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

本页重点：下确界是所有下界中最大的一个。最好先画数轴，再将量词写成 a A 与 ε>0。

读课件时的动作：本页有例子或构造。自己换一个简单数值例子，算出具体集合或函数值；再思考一旦撤掉某条假设，结论是否还成立。

对应的完整证明：参见本讲义最后的「supremum」证明专题。

2.44 第 44 页｜Bounded Monotone Sequence

Bounded Monotone Sequence

Definition. A sequence is bounded if the set of terms from the sequence is bounded.

Definition. A sequence $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ is

(1) increasing if $x_n \leqslant x_{n+1}$ for all $n \in \mathbb{N}$;

(2) decreasing if $x_n \geqslant x_{n+1}$ for all $n \in \mathbb{N}$;

(3) strictly increasing if $x_n < x_{n+1}$ for all $n \in \mathbb{N}$;

(4) strictly decreasing if $x_n > x_{n+1}$ for all $n \in \mathbb{N}$.

A sequence is monotone if it is either increasing or decreasing.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

结合原文理解：上确界证明通常有两步： 证明它是上界； 证明任何更小的数都不是上界。

读课件时的动作：本页以定义为主。建议将英文谓词翻译成“任意…存在…”的中文句子，再试着写一个满足定义和一个不满足定义的例子。

2.45 第 45 页｜Theorem on Limit of Bounded Monotone Sequence

Theorem on Limit of Bounded Monotone Sequence

Theorem on Limit of Bounded Monotone Sequence Every bounded monotone sequence in $\mathbb{R}$ has a limit in $\mathbb{R}$.

Theorem of Limit of Increasing Sequence Suppose $\{x_n : n \in \mathbb{N}\}$ is an increasing sequence of real numbers which is bounded above. Then there exists a limit of $\{x_n : n \in \mathbb{N}\}$, or $\lim_{n\to\infty} x_n$ exists.

Theorem of Limit of Decreasing Sequence Suppose $\{x_n : n \in \mathbb{N}\}$ is a decreasing sequence of real numbers which is bounded below. Then there exists a limit of $\{x_n : n \in \mathbb{N}\}$, or $\lim_{n\to\infty} x_n$ exists.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

结合原文理解：上确界证明通常有两步： 证明它是上界； 证明任何更小的数都不是上界。

读课件时的动作：本页是定理或命题。先明确量词、集合范围、连续性及闭/紧/凸假设；再检查结论是“存在”“唯一”还是“对于任意”。定理本身不能替代证明。

2.46 第 46 页｜本页课件

Theorem on Limit of Bounded Monotone Sequence

Theorem of Limit of Increasing Sequence Suppose $\{x_n : n \in \mathbb{N}\}$ is an increasing sequence of real numbers which is bounded above. Then there exists a limit of $\{x_n : n \in \mathbb{N}\}$, or $\lim_{n\to\infty} x_n$ exists.

Proof. Suppose $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ and $\{x_n : n \in \mathbb{N}\}$ is increasing. Since the set of terms $\{x_n : n \in \mathbb{N}\}$ is bounded above, it has a supremum x. We claim that $\lim_{n\to\infty} x_n = x$.

To see this, note that $x_n \leqslant x$ for all $n \in \mathbb{N}$; but if $\varepsilon > 0$ then there exists $N \in \mathbb{N}$ such that $x_N > x - \varepsilon$, as otherwise $x - \varepsilon$ would be an upper bound. Choose such $N = N(\varepsilon)$. Since $x_N > x - \varepsilon$, then $x_n > x - \varepsilon$ for all $n \geqslant N$ as the sequence is increasing. Hence

$$x - \varepsilon < x_n \leqslant x$$

for all $n \geqslant N$. Thus $|x - x_n| < \varepsilon$ for $n \geqslant N$, and so $\lim_{n\to\infty} x_n = x$ (since $\varepsilon > 0$ is arbitrary).

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

本页重点：上确界不是一定被取到的最大值。验证 s=sup A：所有 a ∈ A 满足 a≤s，并且任意 ε>0 总有 a ∈ A 满足 s−ε<a。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

对应的完整证明：参见本讲义最后的「supremum」证明专题。

2.47 第 47 页｜本页课件

Theorem on Limit of Bounded Monotone Sequence

Theorem of Limit of Decreasing Sequence Suppose $\{x_n : n \in \mathbb{N}\}$ is a decreasing sequence of real numbers which is bounded below. Then there exists a limit of $\{x_n : n \in \mathbb{N}\}$, or $\lim_{n\to\infty} x_n$ exists.

Proof. Suppose $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ and $\{x_n : n \in \mathbb{N}\}$ is decreasing. Since the set of terms $\{x_n : n \in \mathbb{N}\}$ is bounded below, it has a supremum x. We claim that $\lim_{n\to\infty} x_n = x$.

To see this, note that $x_n \geqslant x$ for all $n \in \mathbb{N}$; but if $\varepsilon > 0$ then there exists $N \in \mathbb{N}$ such that $x_N < x + \varepsilon$, as otherwise $x + \varepsilon$ would be a lower bound. Choose such $N = N(\varepsilon)$. Since $x_N < x + \varepsilon$, then $x_n < x + \varepsilon$ for all $n \geqslant N$ as the sequence is increasing. Hence

$$x \leqslant x_n < x + \varepsilon$$

for all $n \geqslant N$. Thus $|x - x_n| < \varepsilon$ for $n \geqslant N$, and so $\lim_{n\to\infty} x_n = x$ (since $\varepsilon > 0$ is arbitrary).

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

本页重点：上确界不是一定被取到的最大值。验证 s=sup A：所有 a ∈ A 满足 a≤s，并且任意 ε>0 总有 a ∈ A 满足 s−ε<a。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

对应的完整证明：参见本讲义最后的「supremum」证明专题。

2.48 第 48 页｜本页课件

Cantor Nested Intervals Theorem

Theorem 1 If $\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals, then there exists a real number $D = sup\{d_n : n \in \mathbb{N}\}$ such that $D \in \bigcap_{n=1}^{\infty} I_n$.

Theorem 2 If $\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals, then there exists a real number $U = inf\{u_n : n \in \mathbb{N}\}$ such that $U \in \bigcap_{n=1}^{\infty} I_n$.

Nested Intervals Theorem If $\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals, then there exist two real numbers $D = sup\{d_n : n \in \mathbb{N}\}$ and $U = inf\{u_n : n \in \mathbb{N}\}$ such that $[D, U] = \bigcap_{n=1}^{\infty} I_n$.

Nested Intervals Theorem If $\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals and $\lim_{n\to\infty} |I_n| = \lim_{n\to\infty} (u_n - d_n) = 0$, then $\lim_{n\to\infty} d_n = \lim_{n\to\infty} u_n \in \bigcap_{n=1}^{\infty} I_n$.

$\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals if $I_1 \supseteq I_2 \supseteq \cdots \supseteq I_n \supseteq \cdots$.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

本页重点：闭区间套的核心是：左端点递增、有上界，取上确界后必须证明它落在每一个区间内。

读课件时的动作：本页是定理或命题。先明确量词、集合范围、连续性及闭/紧/凸假设；再检查结论是“存在”“唯一”还是“对于任意”。定理本身不能替代证明。

对应的完整证明：参见本讲义最后的「nested」证明专题。

2.49 第 49 页｜Cantor Nested Intervals Theorem

Cantor Nested Intervals Theorem

Theorem 1 If $\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals, then there exists a real number $D = sup\{d_n : n \in \mathbb{N}\}$ such that $D \in \bigcap_{n=1}^{\infty} I_n$.

Proof. We note that by the definition of nested intervals that $I_n \subseteq I_1$ for all $n \in \mathbb{N}$ so then $d_n \leqslant u_1$ for all $n \in \mathbb{N}$.
Now consider the non-empty set $A = \{d_n : n \in \mathbb{N}\}$ that is bounded above by u_1. Thus this set has a supremum in the real numbers and denote it $D = \sup A = \lim_{n\to\infty} d_n$ so that $d_n \leqslant D$ for all $n \in \mathbb{N}$.
We now want to show that $D \leqslant u_n$ for all $n \in \mathbb{N}$. We will do this by showing that $d_n \leqslant u_{n'}$ for all $n \in \mathbb{N}$ and $n' \in \mathbb{N}$.
First consider the case where $n \leqslant n'$. We thus have that $I_n \supseteq I_{n'}$ by the definition of nested intervals and so we get that $d_n \leqslant d_{n'} \leqslant u_{n'} \leqslant u_n$. Here we see that $d_n \leqslant u_{n'}$.
Now consider the case where $n > n'$. We thus have that $I_{n'} \supseteq I_n$ by the definition of nested intervals and so we have that $d_{n'} \leqslant d_n \leqslant u_n \leqslant u_{n'}$. Once again we have that $d_n \leqslant u_{n'}$.
So then $d_n \leqslant u_{n'}$ for all $n \in \mathbb{N}$ and $n' \in \mathbb{N}$, and so $u_{n'}$ is an upper bound to the set A and so $D = \sup A \leqslant u_{n'}$ for all $n' \in \mathbb{N}$. Furthermore we have that $d_{n'} \leqslant D$ for all $n' \in \mathbb{N}$, and so $D \in I_{n'}$ for every $n' \in \mathbb{N}$ and thus $D \in \bigcap_{n=1}^{\infty} I_n$ and so the set theoretic union is non-empty.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

本页重点：上确界不是一定被取到的最大值。验证 s=sup A：所有 a ∈ A 满足 a≤s，并且任意 ε>0 总有 a ∈ A 满足 s−ε<a。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

对应的完整证明：参见本讲义最后的「supremum」证明专题。

2.50 第 50 页｜Cantor Nested Intervals Theorem

Cantor Nested Intervals Theorem

Theorem 2 If $\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals, then there exists a real number $U = inf\{u_n : n \in \mathbb{N}\}$ such that $U \in \bigcap_{n=1}^{\infty} I_n$.

Proof. We note that by the definition of nested intervals that $I_n \subseteq I_1$ for all $n \in \mathbb{N}$ so then $d_1 \leqslant u_n$ for all $n \in \mathbb{N}$.
Now consider the non-empty set $B = \{u_n : n \in \mathbb{N}\}$ that is bounded above by d_1. Thus this set has an infumum in the real numbers and denote it $U = \inf B = \lim_{n\to\infty} u_n$ so that $U \leqslant u_n$ for all $n \in \mathbb{N}$.
We now want to show that $I_n \leqslant U$ for all $n \in \mathbb{N}$. We will do this by showing that $d_{n'} \leqslant u_n$ for all $n \in \mathbb{N}$ and $n' \in \mathbb{N}$.
First consider the case where $n \leqslant n'$. We thus have that $I_n \supseteq I_{n'}$ by the definition of nested intervals and so we get that $d_n \leqslant d_{n'} \leqslant u_{n'} \leqslant u_n$. Here we see that $d_{n'} \leqslant u_n$.
Now consider the case where $n > n'$. We thus have that $I_{n'} \supseteq I_n$ by the definition of nested intervals and so we have that $d_{n'} \leqslant d_n \leqslant u_n \leqslant u_{n'}$. Once again we have that $d_{n'} \leqslant u_n$.
So then $d_{n'} \leqslant u_n$ for all $n \in \mathbb{N}$ and $n' \in \mathbb{N}$, and so $I_{n'}$ is a lower bound to the set B and so $u_{n'} \leqslant \sup B = U$ for all $n' \in \mathbb{N}$. Furthermore we have that $U \leqslant u_{n'}$ for all $n' \in \mathbb{N}$, and so $U \in I_{n'}$ for every $n' \in \mathbb{N}$ and thus $U \in \bigcap_{n=1}^{\infty} I_n$ and so the set theoretic union is non-empty.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

本页重点：闭区间套的核心是：左端点递增、有上界，取上确界后必须证明它落在每一个区间内。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

对应的完整证明：参见本讲义最后的「nested」证明专题。

2.51 第 51 页｜本页课件

Cantor Nested Intervals Theorem

Nested Intervals Theorem If $\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals, then there exist two real numbers $D = sup\{d_n : n \in \mathbb{N}\} = \lim_{n\to\infty} d_n$ and $U = inf\{u_n : n \in \mathbb{N}\} = \lim_{n\to\infty} u_n$ such that $[D, U] = \bigcap_{n=1}^{\infty} I_n$.

Proof. We first prove that $[D, U] \subseteq \bigcap_{n=1}^{\infty} I_n$. Now let $x \in [D, U]$, then $D \leqslant x \leqslant U$. But we know that $d_n \leqslant D$ for all $n \in \mathbb{N}$ and $U \leqslant u_n$ for all $n \in \mathbb{N}$ and so $d_n \leqslant D \leqslant U \leqslant u_n$ for all $n \in \mathbb{N}$. Therefore $x \in [d_n, u_n]$ for all $n \in \mathbb{N}$ or in other words $x \in \bigcap_{n=1}^{\infty} I_n$.

Therefore $[D, U] \subseteq \bigcap_{n=1}^{\infty} I_n$.

We then prove that $\bigcap_{n=1}^{\infty} I_n \subseteq [D, U]$. Let $x \in \bigcap_{n=1}^{\infty} I_n$. Then $d_n \leqslant x \leqslant u_n$ for all $n \in \mathbb{N}$. We also know that $d_n \leqslant D \leqslant U \leqslant u_n$ for all $n \in \mathbb{N}$. Suppose that x is such that $d_n \leqslant x < D$. Since D is the supremum of the set A then there exists an element $d_x \in \{d_n : n \in \mathbb{N}\}$ such that $x < d_x$ and so $x \notin [d_x, u_x]$ and therefore $x \notin \bigcap_{n=1}^{\infty} I_n$, a contradiction. Now suppose that x is such that $U < x \leqslant u_n$ for all $n \in \mathbb{N}$. Since U is the infimum of the set B then there exists an element $u_x \in \{u_n : n \in \mathbb{N}\}$ such that $u_x < x$ and so $x \notin [d_x, u_x]$ and so $x \notin \bigcap_{n=1}^{\infty} I_n$, once again, a contradiction. We must therefore have that $d_n \leqslant D \leqslant x \leqslant U \leqslant u_n$ and so $x \in [D, U]$ and so $\bigcap_{n=1}^{\infty} I_n \subseteq [D, U]$.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

本页重点：上确界不是一定被取到的最大值。验证 s=sup A：所有 a ∈ A 满足 a≤s，并且任意 ε>0 总有 a ∈ A 满足 s−ε<a。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

对应的完整证明：参见本讲义最后的「supremum」证明专题。

2.52 第 52 页｜Cantor Nested Intervals Theorem

Cantor Nested Intervals Theorem

Nested Intervals Theorem If $\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals and $\lim_{n\to\infty} |I_n| = \lim_{n\to\infty} (u_n - d_n) = 0$, then $\lim_{n\to\infty} d_n = \lim_{n\to\infty} u_n \in \bigcap_{n=1}^{\infty} I_n$.

Proof. The non-empty set $A = \{d_n : n \in \mathbb{N}\}$ is an increasing sequence of real numbers which is bounded above by u_1, then it has a limit $D = \lim_{n\to\infty} d_n$.
The non-empty set $B = \{u_n : n \in \mathbb{N}\}$ is a decreasing sequence of real numbers which is bounded below by d_1, then it has a limit $U = \lim_{n\to\infty} u_n$.
Thus $d_n \leqslant D \leqslant U \leqslant u_n$ for all $n \in \mathbb{N}$, and

$$0 \leqslant U - D \leqslant u_n - d_n.$$

Note $\lim_{n\to\infty} |I_n| = \lim_{n\to\infty} (u_n - d_n) = 0$, then $D = U \in \bigcap_{n=1}^{\infty} I_n$.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

本页重点：闭区间套的核心是：左端点递增、有上界，取上确界后必须证明它落在每一个区间内。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

对应的完整证明：参见本讲义最后的「nested」证明专题。

2.53 第 53 页｜Bolzano Theorem

Bolzano Theorem

Bolzano Theorem Each bounded sequence in $\mathbb{R}$ has a convergent subsequence.
Bolzano Theorem Each bounded sequence in $\mathbb{R}$ has a limit point.

Proof. Let $\{x_n : n \in \mathbb{N}\}$ be a bounded sequence, we assume $\{x_n : n \in \mathbb{N}\} \subseteq [D, U]$. We divide closed interval $[D, U]$ into two equal-length parts $\left[D, \frac{D+U}{2}\right]$ and $\left[\frac{D+U}{2}, U\right]$. There exists an interval including infinite items of sequence $\{x_n : n \in \mathbb{N}\}$, this interval is written as $[D_1, U_1]$, and we take $x_{n_1} \in [D_1, U_1]$. Then we divide closed interval $[D_1, U_1]$ into two equal-length parts $\left[D_1, \frac{D_1+U_1}{2}\right]$ and $\left[\frac{D_1+U_1}{2}, U_1\right]$. There exists an interval including infinite items of sequence $\{x_n : n \in \mathbb{N}\}$, this interval is written as $[D_2, U_2]$, and we take $x_{n_2} \in [D_2, U_2]$ with $n_2 > n_1$. We repeat this step and obtain a sequence $\{[D_k, U_k] : k \in \mathbb{N}\}$ of closed bounded nested intervals:

$$[D_1, U_1] \supseteq [D_2, U_2] \supseteq \cdots \supseteq [D_k, U_k] \supseteq \cdots \quad \text{with } U_k - D_k = \frac{U - D}{2^k}$$

with a subsequence $\{x_{n_k} : k \in \mathbb{N}\}$ with $x_{n_k} \in [D_k, U_k]$ for $k \in \mathbb{N}$.

From Cantor Nested Intervals Theorem there exists a unique point $a \in \bigcap_{k=1}^{\infty} [D_k, U_k]$, and $|x_{n_k} - a| < U_k - D_k = \frac{U-D}{2^k}$, then $\lim_{k\to\infty} x_{n_k} = a$.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

本页重点：闭区间套的核心是：左端点递增、有上界，取上确界后必须证明它落在每一个区间内。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

对应的完整证明：参见本讲义最后的「nested」证明专题。

2.54 第 54 页｜ Weierstrass Theorem on Accumulation PointWeierstrass Th

Weierstrass Theorem on Accumulation Point

Weierstrass Theorem Every bounded infinite set in $\mathbb{R}$ has an accumulation point.

Proof. Note that E is a bounded infinite set in $\mathbb{R}$, there exists a constant M such that, for any $x \in E$, $|x| \leqslant M$. Define $I_0 = [-M, M]$, we divide R_0 into two small closed intervals: $[-M, 0]$ and $[0, M]$, then at least one small interval includes infinite points in E (otherwise E is a finite set!), we call such a small closed interval I_1 with length M. Generally, we assume that there exists a sequence of closed intervals

$$I_0 \supseteq I_1 \supseteq \cdots \supseteq I_N, \quad I_N \cap E \text{ is an infinite set}$$

where the length of I_n is $\frac{M}{2^{n-1}}$ for $n = 1, 2, \cdots, N$. We divide I_N into two small closed intervals, then at least one small closed interval includes infinite points in E (otherwise E is a finite set!), we call such a small closed interval I_{N+1} with length $\frac{M}{2^N}$. Thus we obtain a sequence of decreasing closed intervals

$$I_0 \supseteq I_1 \supseteq \cdots \supseteq I_n \supseteq \cdots, \quad I_n \cap E \text{ is an infinite set}$$

where the length of I_n is $\frac{M}{2^{n-1}}$ for $n = 1, 2, \cdots$.

这一页怎么读

概念主线： 实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

本页重点： 闭区间套的核心是：左端点递增、有上界，取上确界后必须证明它落在每一个区间内。

读课件时的动作： 本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

对应的完整证明： 参见本讲义最后的「nested」证明专题。

2.55 第 55 页｜Weierstrass Theorem on Accumulation Point

Weierstrass Theorem on Accumulation Point

Weierstrass Theorem Every bounded infinite set in $\mathbb{R}$ has an accumulation point.

Note $\lim_{n\to\infty} \frac{M}{2^{n-1}} = 0$, then, from Cantor Nested Intervals Theorem, there exists a unique point $x \in \bigcap_{n=1}^{\infty} I_n$. We now check that this x is one of accumulation points of E.

Note $\lim_{n\to\infty} \frac{M}{2^{n-1}} = 0$, then for enough large integer n_0, $\frac{M}{2^{n_0}} < \delta$. The length of I_{n_0+1} is strictly less than δ, then $I_{n_0+1} \subseteq N(x,\delta)$, and hence $E \cap I_{n_0+1} \subseteq E \cap N(x,\delta)$. x is an accumulation point of E.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

本页重点：闭区间套的核心是：左端点递增、有上界，取上确界后必须证明它落在每一个区间内。

读课件时的动作：本页是定理或命题。先明确量词、集合范围、连续性及闭/紧/凸假设；再检查结论是“存在”“唯一”还是“对于任意”。定理本身不能替代证明。

对应的完整证明：参见本讲义最后的「nested」证明专题。

2.56 第 56 页｜ Cauchy Convergence Criterion

Cauchy Convergence Criterion

Definition. $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ is a Cauchy sequence if for any $\varepsilon > 0$ there exists an integer N such that, when $n_1 > N$ and $n_2 > N$, $|x_{n_1} - x_{n_2}| < \varepsilon$.

Cauchy Convergence Criterion The sequence $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ is convergent if and only if it is a Cauchy sequence.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

本页重点：柯西条件只比较尾部的各项，而不预设极限；其充分性使用实数的完备性。

读课件时的动作：本页是定理或命题。先明确量词、集合范围、连续性及闭/紧/凸假设；再检查结论是“存在”“唯一”还是“对于任意”。定理本身不能替代证明。

2.57 第 57 页｜ Cauchy Convergence Criterion

Cauchy Convergence Criterion

Cauchy Convergence Criterion The sequence $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ is convergent if and only if it is a Cauchy sequence.

Proof [Necessity] Let $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ be a convergent sequence, and suppose $x = \lim_{n\to\infty} x_n$. For any $\varepsilon > 0$ there exists an integer $N \in \mathbb{N}$ such that, when $n > N$, $|x_n - x| < \varepsilon$. It follows that for any $n_1 > N$ and $n_2 > N$,

$$|x_{n_1} - x_{n_2}| \leqslant |x_{n_1} - x| + |x_{n_2} - x| < 2\varepsilon.$$

Thus $\{x_n : n \in \mathbb{N}\}$ is a Cauchy sequence.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

本页重点：柯西条件只比较尾部的各项，而不预设极限；其充分性使用实数的完备性。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.58 第 58 页｜本页课件

Cauchy Convergence Criterion

[Sufficiency] Let $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ be a Cauchy sequence. For any $\varepsilon > 0$ there exists an integer N such that, when $n_1 > N$ and $n_2 > N$, $|x_{n_1} - x_{n_2}| < \varepsilon$. Then, for any $n > N$, $|x_n - x_{N+1}| < \varepsilon$, or

$$|x_n| \leqslant |x_n - x_{N+1}| + |x_{N+1}| < |x_{N+1}| + \varepsilon \leqslant M$$

where $M = \max\{|x_n| : n = 1, \cdots, N+1\} + \varepsilon$, thus $\{x_n : n \in \mathbb{N}\}$ is bounded.

From Bolzano-Weierstrass Theorem, the bounded sequence $\{x_n : n \in \mathbb{N}\}$ has a convergent subsequence $\{x_{n_k} : k \in \mathbb{N}\}$ with $\lim_{k\to\infty} x_{n_k} = a$. For any $\varepsilon > 0$ there exists an integer K such that, when $k > K$, $|x_{n_k} - a| < \varepsilon$.

Take $k_0 = \max\{K+1, N+1\}$, then $n_{k_0} \geqslant n_{N+1} \geqslant N+1 > N$. Then, when $n > N$, $|x_n - x_{n_{k_0}}| < \varepsilon$, and hence

$$|x_n - a| \leqslant |x_n - x_{n_{k_0}}| + |x_{n_{k_0}} - a| < 2\varepsilon$$

that is, $\lim x_n = a$.

这一页怎么读

概念主线： 实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

本页重点： 有界数列存在收敛子列。反复把区间一分为二，始终留下包含无穷多项的半区间。

读课件时的动作： 本页是定理或命题。先明确量词、集合范围、连续性及闭/紧/凸假设；再检查结论是“存在”“唯一”还是“对于任意”。定理本身不能替代证明。

2.59 第 59 页｜本页课件

Cauchy Convergence Criterion

[Sufficiency] Let $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ be a Cauchy sequence. For any $\varepsilon > 0$ there exists an integer N such that, when $n_1 > N$ and $n_2 > N$, $|x_{n_1} - x_{n_2}| < \varepsilon$. Then, for any $n > N$, $|x_n - x_{N+1}| < \varepsilon$, or

$$|n_n| \leqslant |x_n - x_{N+1}| + |x_{N+1}| < |x_{N+1}| + \varepsilon \leqslant M$$

where $M = \max\{|x_n| : n = 1, \cdots, N+1\} + \varepsilon$, thus $\{x_n : n \in \mathbb{N}\}$ is bounded.

Define $y_n = \inf\{x_n, x_{n+1}, \cdots\}$ for each $n \in \mathbb{N}$. It follows that $y_{n+1} \geqslant y_n$ since y_{n+1} is the infimum over a subset of the set corresponding to y_n, $\{y_n : n \in \mathbb{N}\}$ is an increasing sequence. Moreover the sequence $\{y_n : n \in \mathbb{N}\}$ is bounded since the sequence $\{x_n : n \in \mathbb{N}\}$ is Cauchy and hence bounded. From Theorem on monotone sequences, there exists $a \in \mathbb{R}$ such that $\lim\limits_{n\to\infty} y_n = a$. For any $\varepsilon > 0$ there exists an integer N_1 such that, when $n > N_1$, $|y_n - a| < \frac{1}{2}\varepsilon$, or $a - \frac{1}{2}\varepsilon < y_n < a + \frac{1}{2}\varepsilon$. We claim that $\lim\limits_{n\to\infty} x_n = a$.

这一页怎么读

概念主线： 实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

本页重点： 下确界是所有下界中最大的一个。最好先画数轴，再将量词写成 a A 与 ε>0。

读课件时的动作： 本页是定理或命题。先明确量词、集合范围、连续性及闭/紧/凸假设；再检查结论是“存在”“唯一”还是“对于任意”。定理本身不能替代证明。

对应的完整证明： 参见本讲义最后的「supremum」证明专题。

2.60 第 60 页｜本页课件

Cauchy Convergence Criterion

Let $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ be a Cauchy sequence. For any $\varepsilon > 0$ there exists an integer N_2 such that, when $n_1 > N_2$ and $n_2 > N_2$, $|x_{n_1} - x_{n_2}| < \frac{1}{2}\varepsilon$, or

$$x_{n_1} - \frac{1}{2}\varepsilon < x_{n_2} < x_{n_1} + \frac{1}{2}\varepsilon.$$

From the definition of sequence $\{y_n : n \in \mathbb{N}\}$, $y_n \leqslant x_n$ for each $n \in \mathbb{N}$, then $y_{n_1} - \frac{1}{2}\varepsilon \leqslant x_{n_1} - \frac{1}{2}\varepsilon < x_{n_2}$ for $n_1 > N_1$ and $n_2 > N_2$. On the other hand, $x_{n_2} < x_{n_1} + \frac{1}{2}\varepsilon$, then
$x_{n_2} \leqslant \inf\{x_{n_1} + \frac{1}{2}\varepsilon : n_1 \geqslant N_2\} = \inf\{x_{n_1} : n_1 \geqslant N_2\} + \frac{1}{2}\varepsilon = y_{n_1} + \frac{1}{2}\varepsilon$.
Thus there exists an integer N_2 such that, when $n_1 > N_2$ and $n_2 > N_2$,

$$y_{n_1} - \frac{1}{2}\varepsilon \leqslant x_{n_2} < y_{n_1} + \frac{1}{2}\varepsilon.$$

Take $N = \max\{N_1, N_2\}$, then, when $n_1 > N$ and $n_2 > N$,

$$a - \varepsilon < y_{n_1} - \frac{1}{2}\varepsilon \leqslant x_{n_2} < y_{n_1} + \frac{1}{2}\varepsilon < a + \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, it follows that $\lim\limits_{n\to\infty} x_n = a$.

这一页怎么读

概念主线： 实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

结合原文理解： 上确界证明通常有两步： 证明它是上界； 证明任何更小的数都不是上界。

读课件时的动作： 本页以定义为主。建议将英文谓词翻译成“任意…存在…”的中文句子，再试着写一个满足定义和一个不满足定义的例子。

2.61 第 61 页｜本页课件

Cauchy Convergence Criterion

[Sufficiency] Let $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ be a Cauchy sequence. For any $\varepsilon > 0$ there exists an integer N such that, when $n_1 > N$ and $n_2 > N$, $|x_{n_1} - x_{n_2}| < \varepsilon$. Then, for any $n > N$, $|x_n - x_{N+1}| < \varepsilon$, or

$$|n_n| \leqslant |x_n - x_{N+1}| + |x_{N+1}| < |x_{N+1}| + \varepsilon \leqslant M$$

where $M = \max\{|x_n| : n = 1, \cdots, N+1\} + \varepsilon$, thus $\{x_n : n \in \mathbb{N}\}$ is bounded.

Define $y_n = \sup\{x_n, x_{n+1}, \cdots\}$ for each $n \in \mathbb{N}$. It follows that $y_{n+1} \leqslant y_n$ since y_{n+1} is the supremum over a subset of the set corresponding to y_n, $\{y_n : n \in \mathbb{N}\}$ is a decreasing sequence. Moreover the sequence $\{y_n : n \in \mathbb{N}\}$ is bounded since the sequence $\{x_n : n \in \mathbb{N}\}$ is Cauchy and hence bounded. From Theorem on monotone sequences, there exists $a \in \mathbb{R}$ such that $\lim\limits_{n\to\infty} y_n = a$. For any $\varepsilon > 0$ there exists an integer N_1 such that, when $n > N_1$, $|y_n - a| < \frac{1}{2}\varepsilon$, or $a - \frac{1}{2}\varepsilon < y_n < a + \frac{1}{2}\varepsilon$. We claim that $\lim\limits_{n\to\infty} x_n = a$.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

本页重点：上确界不是一定被取到的最大值。验证 s=sup A：所有 a ∈ A 满足 a≤s，并且任意 ε>0 总有 a ∈ A 满足 s−ε<a。

读课件时的动作：本页是定理或命题。先明确量词、集合范围、连续性及闭/紧/凸假设；再检查结论是“存在”“唯一”还是“对于任意”。定理本身不能替代证明。

对应的完整证明：参见本讲义最后的「supremum」证明专题。

2.62 第 62 页｜ Cauchy Convergence CriterionLet {xn : n N} R be a Cau

Cauchy Convergence Criterion

Let $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ be a Cauchy sequence. For any $\varepsilon > 0$ there exists an integer N_2 such that, when $n_1 > N_2$ and $n_2 > N_2$, $|x_{n_1} - x_{n_2}| < \frac{1}{2}\varepsilon$, or

$$x_{n_1} - \frac{1}{2}\varepsilon < x_{n_2} < x_{n_1} + \frac{1}{2}\varepsilon.$$

From the definition of sequence $\{y_n : n \in \mathbb{N}\}$, $y_n \geqslant x_n$ for each $n \in \mathbb{N}$, then $x_{n_2} < x_{n_1} + \frac{1}{2}\varepsilon \leqslant y_{n_1} + \frac{1}{2}\varepsilon$ for $n_1 > N_1$ and $n_2 > N_2$. On the other hand, $x_{n_1} - \frac{1}{2}\varepsilon < x_{n_2}$, then
$y_{n_1} - \frac{1}{2}\varepsilon = \sup\{x_{n_1} : n_1 \geqslant N_2\} - \frac{1}{2}\varepsilon = \sup\{x_{n_1} - \frac{1}{2}\varepsilon : n_1 \geqslant N_2\} \leqslant x_{n_2}$. Thus there exists an integer N_2 such that, when $n_1 > N_2$ and $n_2 > N_2$,

$$y_{n_1} - \frac{1}{2}\varepsilon < x_{n_2} \leqslant y_{n_1} + \frac{1}{2}\varepsilon.$$

Take $N = \max\{N_1, N_2\}$, then, when $n_1 > N$ and $n_2 > N$,

$$a - \varepsilon < y_{n_1} - \frac{1}{2}\varepsilon < x_{n_2} \leqslant y_{n_1} + \frac{1}{2}\varepsilon < a + \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, it follows that $\lim\limits_{n\to\infty} x_n = a$.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

本页重点：柯西条件只比较尾部的各项，而不预设极限；其充分性使用实数的完备性。

读课件时的动作：本页以定义为主。建议将英文谓词翻译成“任意…存在…”的中文句子，再试着写一个满足定义和一个不满足定义的例子。

2.63 第 63 页｜ Heine - Borel Theorem on Finite Coverings

Heine - Borel Theorem on Finite Coverings

A collection $\mathscr{H}$ of open sets is an open covering of a set S if every point in S is contained in a set H belonging to $\mathscr{H}$; that is, $S \subseteq \{H \in \mathscr{H}\}$.

Heine - Borel Theorem If $\mathscr{H}$ is an open covering of a closed and bounded subset S of the real line; then S has an open covering $\tilde{\mathscr{H}}$ consisting of finitely many open sets belonging to $\mathscr{H}$.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

本页重点：开覆盖具有有限子覆盖的结论必须依赖紧致性。证明时可用闭区间套反证或逐次细分。

读课件时的动作：本页是定理或命题。先明确量词、集合范围、连续性及闭/紧/凸假设；再检查结论是“存在”“唯一”还是“对于任意”。定理本身不能替代证明。

2.64 第 64 页｜Heine - Borel Theorem on Finite Coverings

Heine - Borel Theorem on Finite Coverings

Heine - Borel Theorem If $\mathscr{H}$ is an open covering of a closed and bounded subset S of the real line; then S has an open covering $\tilde{\mathscr{H}}$ consisting of finitely many open sets belonging to $\mathscr{H}$.

Proof. (Reduction to Absurdity) Let $\mathscr{H}$ be an open covering of $[D, U]$. Assume that $[D, U]$ can not be covered by finite many open sets belonging to $\mathscr{H}$. We divide closed interval $[D, U]$ into two equal-length parts $\left[D, \frac{D+U}{2}\right]$ and $\left[\frac{D+U}{2}, U\right]$. There exists at least one small interval that can not be covered by finite many open sets belonging to $\mathscr{H}$, this interval is written as $[D_1, U_1]$. Then we divide closed interval $[D_1, U_1]$ into two equal-length parts, then there exists at least one small interval that can not be covered by finite many open sets belonging to $\mathscr{H}$, this interval is written as $[D_2, U_2]$. We repeat this step and obtain a sequence $\{[D_n, U_n] : n \in \mathbb{N}\}$ of closed bounded nested intervals that can not be covered by finite many open sets belonging to $\mathscr{H}$:

$$[D_1, U_1] \supseteq [D_2, U_2] \supseteq \cdots \supseteq [D_n, U_n] \supseteq \cdots \quad \text{with } U_n - D_n = \frac{U - D}{2^n}.$$

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

本页重点：闭区间套的核心是：左端点递增、有上界，取上确界后必须证明它落在每一个区间内。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

对应的完整证明：参见本讲义最后的「nested」证明专题。

2.65 第 65 页｜Heine - Borel Theorem on Finite Coverings

Heine - Borel Theorem on Finite Coverings

Heine - Borel Theorem If $\mathscr{H}$ is an open covering of a closed and bounded subset S of the real line; then S has an open covering $\tilde{\mathscr{H}}$ consisting of finitely many open sets belonging to $\mathscr{H}$.

From Cantor Nested Intervals Theorem there exists a unique point $x \in \bigcap_{n=1}^{\infty} [D_n, U_n]$. Note that $\mathscr{H}$ is an open covering of $[D, U]$, then there exists $H_0 \in \mathscr{H}$ such that $x \in H_0$. Note that H_0 is an open set, then there exists $\delta > 0$ such that $(x - \delta, x + \delta) \subseteq H_0$.

Note $\lim_{n\to\infty} D_n = x = \lim_{n\to\infty} U_n$, then there exists an integer N such that, when $n > N$, $D_n \in (x - \delta, x + \delta)$ and $U_n \in (x - \delta, x + \delta)$.

$$[D_n, U_n] \subseteq (x - \delta, x + \delta) \subseteq H_0 \qquad \text{for } n > N.$$

which contradicts that $[D_n, U_n]$ can not be covered by finite many open sets belonging to $\mathscr{H}$.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

本页重点：闭区间套的核心是：左端点递增、有上界，取上确界后必须证明它落在每一个区间内。

读课件时的动作：本页是定理或命题。先明确量词、集合范围、连续性及闭/紧/凸假设；再检查结论是“存在”“唯一”还是“对于任意”。定理本身不能替代证明。

对应的完整证明：参见本讲义最后的「nested」证明专题。

2.66 第 66 页｜Real Number Space R

Real Number Space ℝ

Dedekind - Cantor Axiom of Continuity of a Real Line Suppose the pair of sets (A, A^c) is a Dedekind Cut of $\mathbb{R}$ with the property: if $a \in A$ and $b \in A^c$, then $a < b$, then either there exists a maximum in A or there exists a minimum in A^c.

Theorem of Supremum and Infimum Suppose E is a non-empty set of real numbers which is bounded above, then E has a supremum in $\mathbb{R}$. Suppose E is a non-empty set of real numbers which is bounded below, then E has a infimum in $\mathbb{R}$.

Theorem of Limit of Monotone Sequence Suppose $\{x_n : n \in \mathbb{N}\}$ is an increasing sequence of real numbers which is bounded above, then there exists a limit of $\{x_n : n \in \mathbb{N}\}$, or $\lim_{n\to\infty} x_n$ exists. Suppose $\{x_n : n \in \mathbb{N}\}$ is a decreasing sequence of real numbers which is bounded below, then there exists a limit of $\{x_n : n \in \mathbb{N}\}$, or $\lim_{n\to\infty} x_n$ exists.

Nested Intervals Theorem If the interval $\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals, then there exist two real numbers $D = sup\{d_n : n \in \mathbb{N}\}$ and $U = inf\{u_n : n \in \mathbb{N}\}$ such that $[D, U] \subset \cap_{n=1}^{\infty} I_n$.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

本页重点：上确界不是一定被取到的最大值。验证 s=sup A：所有 a ∈ A 满足 a≤s，并且任意 ε>0 总有 a ∈ A 满足 s−ε<a。

读课件时的动作：本页是定理或命题。先明确量词、集合范围、连续性及闭/紧/凸假设；再检查结论是“存在”“唯一”还是“对于任意”。定理本身不能替代证明。

对应的完整证明：参见本讲义最后的「supremum」证明专题。

2.67 第 67 页｜ Real Number Space R

Real Number Space $\mathbb{R}$

Bolzano Theorem Each bounded sequence in $\mathbb{R}$ has a convergent subsequence.
Weierstrass Theorem Every bounded infinite set in $\mathbb{R}$ has an accumulation point.
Cauchy Convergence Criterion The sequence $\{x_n : n \in \mathbb{N}\}$ in $\mathbb{R}$ is convergent if and only if it is a Cauchy sequence: for any $\varepsilon > 0$, there exist a large natural number N such that, when $n_1 > N$ and $n_2 > N$, $|x_{n_1} - x_{n_2}| < \varepsilon$.
Heine - Borel Theorem If $\mathscr{H}$ is an open covering of a closed and bounded subset E of the real line; then E has an open covering $\tilde{\mathscr{H}}$ consisting of finitely many open sets belonging to $\mathscr{H}$.

这一页怎么读

概念主线：实数的核心是有序域加完备性。最小上界不是简单的“最大元素”；上确界可以不属于原集合。

本页重点：有界数列存在收敛子列。反复把区间一分为二，始终留下包含无穷多项的半区间。

读课件时的动作：本页是定理或命题。先明确量词、集合范围、连续性及闭/紧/凸假设；再检查结论是“存在”“唯一”还是“对于任意”。定理本身不能替代证明。

3 证明专题：考试应会的标准推导

专题的使用方法

这些专题沿着本课 PPT 的定理顺序补足主要推导；若原课件的命题另有附加假设，以该页原文为主，切勿将证明中的条件删掉。对于高维拓扑不动点定理，只陈述严格的适用条件并说明所需工具，不伪称给出了没有依据的完整高维证明。

3.1 专题：complete

证明｜为什么有理数不是完备的

考虑 $A=\{q\in\mathbb{Q}:q>0,\ q^2<2\}$。它在 $\mathbb{Q}$ 内非空、有上界，却没有有理数上确界。若 $s\in\mathbb{Q}$ 是其上确界，则 $s^2=2$ 才可能；若 $s^2<2$，可以在 s 附近找更大的有理数仍满足平方小于 2；若 $s^2>2$，可以找一个稍小的有理数仍是上界。这两种情况均与上确界矛盾。又由初等数论 $\sqrt{2}\notin\mathbb{Q}$（假设 $\sqrt{2}=a/b$ 为最简分数，可推出 a,b 均为偶数），故 $\mathbb{Q}$ 不完备。

3.2 专题：supremum

证明｜上确界的 ε 刻画

设 $A\subset\mathbb{R}$ 非空且有上界，$s=\sup A$。第一，由定义，对任意 $a\in A$ 有 $a\le s$。第二，对任意 $\varepsilon>0$，若不存在 $a>s-\varepsilon$，则 $s-\varepsilon$ 也是上界，与 s 为最小上界矛盾。所以 $\exists a\in A$ 使 $s-\varepsilon<a\le s$。反过来，若 s 是上界且满足上述逼近性，假设有更小上界 $b<s$，取 $\varepsilon=s-b$，找到 $a>b$，矛盾。

3.3 专题：archimedean

证明｜由最小上界性质证明阿基米德性

任取 $x\in\mathbb{R}$。若对每个 $n\in\mathbb{N}$ 都有 $n\le x$，则 $\mathbb{N}$ 有上界，令 $s=\sup\mathbb{N}$。因为 $s-1$ 不是上界，存在 $n>s-1$，于是 $n+1>s$。但 $n+1\in\mathbb{N}$，与 s 是上界矛盾。故存在 $n\in\mathbb{N}$ 使 $n>x$。

3.4 专题：nested

证明｜闭区间套定理（完整量词）

设 $I_n=[a_n,b_n]$ 为一列非空闭区间，$I_{n+1}\subseteq I_n$。则 a_n 单调不减、b_n 单调不增，且 $a_n\le b_m$ 对所有 m,n 成立（取足够大的 $k\ge m,n$ 可得 $a_n\le a_k\le b_k\le b_m$）。集合 $A=\{a_n:n\ge1\}$ 非空、有上界，令 $c=\sup A$。对任意固定 m，b_m 是 A 的上界，故 $c\le b_m$；同时 $a_m\le c$。因此 $a_m\le c\le b_m$，从而 $c\in\bigcap_n I_n$。若再有 $b_n-a_n\to0$，任意 c,d 在交集中都满足 $|c-d|\le b_n-a_n\to0$，故 $c=d$，交集恰有一个点。

4 自测清单

1. 你能不看课件写出本讲的核心定义及所有量词吗?
2. 你能逐条列出每一个定理的假设，并解释删去一条为何可能失败吗?
3. 你能在白纸上独立复述本讲证明专题中的每一个箭头吗?
4. 你能结合原课件的反例判断是否把充分条件错写成必要条件吗?